Online Resource 1

PlanArena: an exact, learning-assisted land-use optimizer with empirical evaluation
Computational Urban Science. Model specification, test protocols and additional results.
Author details are withheld for double-blind review.

This supplement specifies every model in the paper and the protocol of every test, so that each result can be reproduced from the code and the public data. Settings marked † are read from the code when this document is built. Section numbers refer to the main text unless prefixed with S.

S1 Types of evidence

The paper uses four kinds of evidence, which support different claims (Table S1).

- **Empirical predictive validation**: a prediction made with information available before an outcome, compared with that outcome, on observations not used to build or select the model.
- **Accounting check**: the model reproduces recorded quantities it was built from or calibrated to (for example the baseline twin against assessor records). It shows the bookkeeping is right, not that predictions are.
- **Algorithmic verification**: a search or rule is checked against a known answer (exhaustive enumeration, exact re-scoring), independent of whether its objective is right.
- **Illustrative simulation**: outputs that follow from stated assumptions (stakeholder utilities, seats) and show how a decision depends on them. They have no observed counterpart.

S2 Data timing and leakage audit

Feature timestamps. Table S2 lists the source and date of every feature family for each panel. In Manhattan every feature comes from the MapPLUTO release of the decision year t_0. In the cross-city panels, zoning and plan layers are current layers in some cities; lots whose zoning or plan dates from t_0 or later are flagged where the layer carries dates, and the conservative variant drops them. Dallas's zoning and plan are undated current layers (the plan was adopted in 2024, after its t_0 of 2022), so its zoning and plan results may contain post-decision information and are not unbiased estimates of what those layers add.

Following lots over time. Each t_0 lot is followed to the t_1 lot that contains its representative point. When several t_0 lots fall in one t_1 lot (a merger), their t_0 floor area is summed before growth is measured, and each of them takes the merged lot's outcome. When one t_0 lot contains several t_1 lots (a split), growth is measured on the piece that contains the t_0 point. Table S3 counts both. Lots with no t_1 lot are left out.

A label error found and corrected. A building counts as new in $(t_0, t_1]$ only if it is not already on the t_0 record with a year built of t_0 or later. The December 2020 MapPLUTO release (20v8) already lists 50 buildings completed in 2020; without this rule they were counted as developed in every pair starting in 2020 (78% of the 2020–2021 positives and 9% of the 2020–2025 positives). Earlier releases are dated early in the year and are unaffected. The same rule was added to the Los Angeles and Dallas panels (0.8% of positives at most). All results in the paper use the corrected labels.

Separate observations. In the held-out impact test (Section S4), the delta form, the unit-size rule and the scale factors are chosen on lots in two of five spatial folds, and all metrics are computed on the other three. Coefficients are fitted on data from before the changes (benchmarking and LEHD counts of 2015). In the likelihood tests the model is fitted only on pairs whose outcome was known at the decision year. Hyperparameters are fixed in the code and were not tuned on test periods; the graph neural network's configuration was selected on a validation split of the training period.

S3 Impact engine

Notation. Parcel i has lot area A_i, existing use u_i^0 in one of nine analysis groups (residential, mixed, commercial, industrial, public facility, open space, parking, vacant, transportation and utility), floor area F_i^0, homes H_i^0, assessed

Table S1 Claims, the evidence behind them, and its type.

Component	Claim	Evidence	Test
Existing conditions	records reproduced	accounting check	twin vs assessor, LL84, LEHD
Planning calculations	reviews' projections reproduced	accounting check	two environmental reviews
Impact engine	effects of observed change	empirical, held out	selection / evaluation folds
Job change	effect on jobs by block	empirical, held out (no skill)	LEHD 2015–2023
Access, mix, green cover	effects of change	not validated	no after-change data
Redevelopment model	where change happens	empirical, out of time	rolling origins; seven cities
Calibration	how often change happens	empirical, out of time	reliability, O/E, Brier skill
Zoning and plan layers	added predictive value	empirical, out of time	block ablation, conservative variant
Housing projections	homes built	empirical, out of time	districts; three rezonings
Optimizer	best plan found	algorithmic verification	exhaustive search; benchmark
NSGA-III explorer	trade-off front	algorithmic verification	exhaustive enumeration of blocks
Realism layers	plausible recommendations	empirical, out of time	2020 plans vs 2025 outcomes
Decision module	how a choice depends on its rule	illustrative simulation	assumed utilities and seats

Table S2 Source and date of each feature family against the decision year t_0. Context features (neighbour composition, recent construction within 400 m) come from the same record as existing conditions.

Panel	Existing conditions	Zoning	Plan	Measured after t_0
Manhattan (rolling origins, 2015-2025)	MapPLUTO release of year t0: use, lot and floor area, floors, year built, units, assessed land and total value	same release: zoning district, maximum FARs, overlays, landmark and historic status	none (no citywide adopted land-use plan)	none
Los Angeles	SCAG 2020 land use; assessor roll of t0	SCAG 2019 compilation	SCAG 2019 General Plan compilation	none
Dallas	DCAD codes of t0	current layer (undated)	forwardDallas! 2.0, adopted 2024 (after t0 = 2022)	zoning and plan may include changes after t0; no dates to remove them
Phoenix	MAG vintage of t0; assessor	current layer; zoning cases dated t0 or later flagged	General Plan 2015 as amended; polygons added after t0 flagged	flagged lots removed in the conservative variant
San Diego	SANDAG vintage of t0	current layer; implemented on or after t0 flagged	General and community plans (undated)	zoning flags removed in the conservative variant; plan undated
New York City	MapPLUTO of t0	MapPLUTO of t0	none	none
Milwaukee	MPROP of t0	MPROP of t0	none (PDF only)	none
Chicago	CMAP inventory before t0; Cook County Assessor of t0	current layer; districts created on or after t0 flagged	none	flagged lots removed in the conservative variant

value V_i^0 and floor-area components c_{ik}^0 for $k \in$ {residential, office, retail, other commercial, factory, storage, public facility, transport facility}, all from the assessor or city record. A plan sets new uses u_i and, optionally, floor-area ratios f_i.

Floor area of a changed parcel.

$$F_i = \begin{cases} A_i f_i & \text{if the planner sets a FAR,} \\ F_i^0 & \text{if a building exists and the new use is built (reuse),} \\ A_i \, \bar{f}_{iu} \, \rho & \text{if the lot is unbuilt,} \end{cases} \tag{S1}$$

where $\bar{f}_{iu}$ is the zoning maximum FAR for the use and $\rho = 0.8^{\dagger}$ the build-out utilisation (range 0.6–1.0). Modelled components $m_{ik}(u, F)$ split F by use: mixed use by the median residential share of mixed-use buildings, commercial by the city's office/retail/other shares, industrial by its factory/storage shares.

Delta forms. Unchanged parcels keep their recorded values, so the baseline reproduces the records. For a changed parcel, three forms were tested:

$$\text{replacement:} \quad c_{ik} = m_{ik}(u_i, F_i), \qquad H_i = m_{i,\text{res}}/q_i; \tag{S2}$$

$$\text{difference:} \quad c_{ik} = \max\{0,\ c_{ik}^0 + m_{ik}(u_i, F_i) - m_{ik}(u_i^0, F_i^0)\}; \tag{S3}$$

$$\text{hybrid:} \quad c_{ik} = c_{ik}^0 F_i / F_i^0,\ H_i = H_i^0 F_i / F_i^0 \text{ if } u_i = u_i^0 \text{ and the record is consistent; else replacement.} \tag{S4}$$

Table S3 Following Manhattan lots over time, and labels removed by the new-building rule.

Pair	t_0 lots	No t_1 lot	Mergers	Lots merged	Splits	Developed	Removed
2015–2016	42,786	0	92	239	24	913	1
2015–2020	42,786	1	307	872	55	1,814	1
2019–2020	42,441	1	60	182	10	313	0
2020–2021	42,327	0	4	11	2	14	51
2020–2025	42,327	1	153	472	43	520	50
2024–2025	42,134	0	38	115	14	126	0

A record is consistent if $F_i^0 \geq 93$ m^2 (1,000 sq ft) and, if it has homes, $23 \leq F_i^0/H_i^0 \leq 465$ m^2 (250–5,000 sq ft). The unit size q_i is the parcel's own floor area per home when it lies in that range, and otherwise the city median ($85^\dagger$ m^2, or 913 sq ft, in Manhattan). The deployed engine uses the hybrid form.

Derived quantities.

$$\text{jobs:} \quad J_i = \textstyle\sum_k c_{ik}/\phi_k, \tag{S5}$$

$$\text{emissions:} \quad G_i = F_i \,\mathrm{EUI}_{u_i}\,\mathrm{EF}_{u_i}, \tag{S6}$$

$$\text{runoff:} \quad R_i = A_i\,[\kappa_i C_{\text{roof}} + (1-\kappa_i)\,C_{u_i}], \tag{S7}$$

$$\text{value:} \quad V_i = \max\{0,\ V_i^0 + v(u_i,F_i) - v(u_i^0,F_i^0)\}, \quad v(u,F) = A_i\,\ell_d + F\,b_{u,d}, \tag{S8}$$

with floor area per worker ϕ_k scaled so that baseline jobs match the LEHD total; energy-use intensity and emission factor by use from New York's benchmarking data (Local Law 84) and emission limits (Local Law 97); roof share κ_i from floor area over floors; rational-method coefficients C; and median land value per unit lot area ℓ_d and building value per unit floor area $b_{u,d}$ of the community district. Green space is the lot area of open space. Residents are homes times $1.95^\dagger$ persons per household.

Access, mix and green cover. Parcels are rasterised on a 50 m† grid. Access uses Euclidean disks, not walking routes: radius 400 m† (about five minutes) and 800 m† (about ten). Each indicator is a mean over residents in the scope:

- **park access**: share of residents with at least 0.4 ha† (1 acre) of open space within the short radius;
- **services and facility adequacy**: $\min\{1, S_i/(P_i\,t)\}$, where S_i is retail and other commercial (or public-facility) floor area within the radius, P_i the residents within it, and t the city's median per-resident value at baseline;
- **land-use mix**: normalised entropy of the lot area of five land groups within the short radius;
- **green cover**: the open-space share of that land.

Scores, pillars and overall score. Let b_k and s_k be indicator k's baseline and scenario values in the scope (the ten-minute disk around the changed parcels, a district or the city), and $\delta_k = \pm 1$ its direction.

$$\text{score}_k = \text{clip}_{[-100,100]} \begin{cases} 100\,\delta_k(s_k - b_k)/|b_k| & \text{totals (per cent change)} \\ 100\,\delta_k(s_k - b_k) & \text{shares (percentage points)} \end{cases} \tag{S9}$$

A pillar score is the mean of its indicators' scores (economic: homes, jobs, value; social: park, services, facility access and mix; environmental: green space, runoff, emissions and green cover). The overall score is $\sum_p w_p\,\text{pillar}_p / \sum_p w_p$ with planner-set weights w_p, equal by default. Because totals are scored relative to the scope's baseline, a change scores higher in a smaller scope; comparisons are made within one scope.

Sensitivity ranges. Each coefficient has a central value and a low and high value from its source (sources and status are recorded with each coefficient in `planarena/indicators/coefficients.yaml`). Two variants set every ranged coefficient to the end that lowers (variant a) or raises (variant b) the quantity, together with the 25th and 75th percentiles of building value per unit floor area and of floor area per home. The range shown for an indicator is the minimum and maximum of the three deltas (central, a, b). It is a sensitivity range: it has no probability attached and does not include model error, so it is not a prediction interval.

Consistency checks (tests in the repository): the baseline scenario gives zero deltas on every indicator; undoing an edit restores the baseline exactly; scores stay in $[-100, 100]$; baseline jobs equal the calibration total.

Table S4 Held-out impact test on 2,036 evaluation lots: Pearson r, mean absolute error (homes), mean bias (predicted minus observed) and total predicted over total observed.

Method	Homes r	MAE	Bias	Total	Value r	Total	Emis. r	Total	n emis.
PlanArena (form chosen on selection lots)	0.79	12.7	6.1	1.51	0.74	0.70	0.60	1.24	115
× selection-lot scale	0.79	11.8	4.2	1.35	0.74	1.09	0.60	1.06	115
PlanArena (deployed form)	0.79	12.5	5.6	1.47	0.74	0.70	0.60	1.24	115
Mean change of the same transition	0.39	22.3	1.5	1.12	0.10	1.07	0.26	2.26	115
Gradient-boosted regression	0.68	15.4	2.9	1.24	0.57	1.19	0.36	1.10	115

Predicting no change gives a homes MAE of 16.5. Bias and MAE in homes per lot.

Table S5 Deployed engine by size of the new building (m^2) and by kind of change (evaluation lots).

Group	n	Homes r	MAE	Bias	Total	Value total
$<$929 m^2	882	0.90	1.7	1.1	−1.22	0.69
10k-50k	687	0.75	6.3	3.0	2.11	0.90
50k-200k	360	0.79	21.1	5.6	1.27	0.99
$\geq$18,581 m^2	107	0.73	113.2	59.9	1.41	0.58
use change	1,964	0.83	11.6	4.8	1.40	0.67
density only (new building)	72	−0.24	37.9	28.9	3.25	0.97

Total: predicted over observed total; negative where the two totals have opposite signs (small buildings lost homes in total).

S4 Held-out test of the impact engine

Design. 3,350 Manhattan lots changed use or floor area substantially between 2015 and 2025 (new buildings and use changes of at least 186 m^2, or 2,000 sq ft). Each is given its 2015 state and its observed 2025 use and floor area, and the engine predicts the change in homes, assessed value and emissions; observed changes come from the 2025 record (homes, value) and from benchmarked buildings in both years (emissions). The test is conditional: it measures the engine's arithmetic given what was built, not whether the change would happen. The lots are split into five spatial folds. On folds 0–1 (1,314 lots) the delta form and unit-size rule are chosen by mean normalised absolute error over the three indicators, and one scale factor per indicator (total observed over total predicted) is fitted. Folds 2–4 (2,036 lots) are used only for evaluation. Coefficients come from 2015 data (benchmarking 2015, LEHD 2015), before the changes.

Baselines, fitted on the selection folds with the same information (old and new use, floor area, lot area, old homes and value): the mean change of the same transition and size class, and gradient-boosted regression.

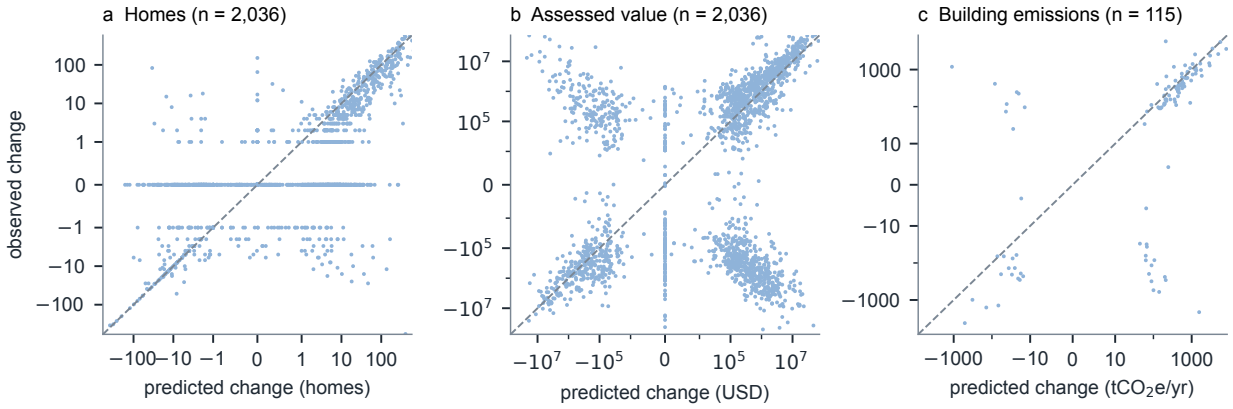


Fig. S1 Observed against predicted change on the evaluation lots (symmetric log axes; dashed line: equality)

Direction of change. Among lots where both changes are non-zero, the predicted change has the observed sign for 88% of homes, 60% of assessed-value changes and 80% of emissions changes (Figure S1). The correlation for value is carried by large lots; for a single lot, the direction of the value change is unreliable.

Jobs. Job change was tested by 2020 census block (LEHD 2015 and 2023, adjusted for citywide growth), on blocks with no selection-fold lot: $r = 0.05$ over 837 blocks. The engine has no skill for job change at this scale, and the Studio marks job change as not validated.

Table S6 Chosen engine form by transition (groups with at least 30 evaluation lots).

Transition	n	Homes MAE	Bias	Total	Value total
res → mix	372	9.5	−3.2	1.96	0.61
ind → mix	239	7.5	7.0	2.19	0.63
ind → com	235	0.6	−0.6	2.74	1.61
com → mix	224	30.8	27.9	2.13	0.32
mix → res	112	5.7	3.9	3.88	−0.43
vac → opn	100	0.0	0.0	–	0.00
mix → com	81	1.0	−1.0	1.86	1.19
com → vac	43	0.0	0.0	1.00	1.10
vac → mix	42	29.8	20.6	1.30	0.77
prk → mix	40	68.9	−8.5	0.95	0.69
vac → res	33	15.7	4.3	1.12	0.48
mix → mix	31	49.2	45.7	2.28	0.70

Table S7 Learned correction of the engine and calibrated intervals on the held-out lots (2015–2025 changes; engine with pre-period coefficients).

	Lots	MAE	MAE+	Bias	Bias+	r	r+	Total	Total+	Adopted	90% int.	Range
Homes	2,036	12.5	10.6	5.6	3.9	0.79	0.80	1.47	1.32	yes	88%	66%
Value (USD 1,000)	2,036	2,994	2,643	−682	−293	0.74	0.75	0.70	0.87	yes	85%	19%
Emissions (tCO_2e)	115	519	531	123	−63	0.60	0.61	1.24	0.87	no	98%	38%

+: engine plus the learned correction. Total: predicted over observed total. 90% int.: share of held-out lots whose observed change lies in the conformal interval (nominal 90%). Range: share inside the engine's low–high sensitivity range.

Learned correction and calibrated intervals. On the same events, folds and pre-period coefficients, a gradient-boosted model (LightGBM) of the engine's error uses the engine's prediction and the baselines' lot information. Its loss (squared or Huber) is chosen by cross-fitting between the two selection folds; it is trained on the selection folds and evaluated on the held-out folds, and it is adopted only if it lowers the held-out mean absolute error and absolute bias without lowering the correlation. Prediction intervals come from conformalized quantile regression of the error (Romano et al., 2019): quantile models (5th and 95th percentiles) fitted on one selection fold and calibrated on the other, for a nominal 90% coverage. Table S7 compares them with the engine and with how often the engine's sensitivity range contains the observed change. The Studio's parcel inspector shows the adopted corrections and their intervals (refitted on all events, calibrated on the held-out folds) beside the engine's yields; the engine's own scores are unchanged.

S5 Likelihood, yield and calibration

Outcome. A lot is developed in $(t_0, t_1]$ if a new building appears on it (Section S2) or its floor area grows by more than 25% and 186 m^2 (2,000 sq ft). Cross-city definitions follow each source (main text, Sect. 4).

Manhattan model. LightGBM (500 trees, learning rate 0.03, 31 leaves, at least 40 lots per leaf, row and column subsampling 0.8), fitted on all pairs $(a, a+h)$ with $a+h \le t_0$, on 59 features from the t_0 record: lot and floor area, floors, age, built and permitted FARs, overlay, landmark and historic status, zoning family, existing use, unused permitted floor area, land share of value, value per unit lot area, soft-site flag, homes, neighbour use composition on the parcel graph, and the share of neighbours built in the five years before t_0.

Cross-city model. LightGBM (400 trees, learning rate 0.04, 31 leaves, at least 50 lots per leaf, subsampling 0.8) on the blocks E, C, Z, P (main text, Sect. 4), fitted on the first window and tested on the second.

Calibration metrics. For probabilities p and outcomes y in a test window: reliability in ten equal-count bins; expected calibration error (bin-weighted absolute gap); observed over predicted events $\sum y / \sum p$; calibration intercept and slope from a logistic regression of y on $\mathrm{logit}(p)$ (ideal 0 and 1; a slope below 1 means the probabilities are too spread out); and the Brier score against a constant forecast of the training-period rate, as skill $1 - \mathrm{Brier}/\mathrm{Brier}_{\mathrm{prevalence}}$.

Recalibration, using training data only, in two ways: *isotonic*, fitted on the latest training pair as predicted by a model trained on the earlier pairs (Manhattan, where several pairs exist); and *logistic* (intercept and slope on logit p), fitted on out-of-fold probabilities of the training data from five spatial folds. Neither uses the test window.

From probability to homes. Three projections are compared with homes built.

- **Capacity**: each site built to the permitted floor area at utilisation ρ, homes K_i.

Table S8 Calibration of the Manhattan redevelopment model by horizon h (years) and test window: observed over predicted events (O/E), calibration intercept and slope, and Brier skill against the training-period rate; then after temporal isotonic recalibration, and after logistic recalibration on out-of-fold training probabilities.

h	Test	Lots	Events	O/E	Int.	Slope	Skill	O/E iso	Skill iso	O/E rec.	Slope rec.	Skill rec.
1	2016–2017	42,524	183	0.98	−1.31	0.65	0.035	–	–	0.43	0.89	0.008
1	2017–2018	42,491	204	1.28	−0.56	0.78	0.113	1.17	0.104	0.74	0.99	0.118
1	2018–2019	42,431	267	1.28	−0.42	0.80	0.154	1.04	0.153	0.97	0.97	0.160
1	2019–2020	42,307	313	1.70	0.15	0.88	0.105	1.18	0.097	1.31	1.04	0.105
1	2020–2021	42,171	14	0.07	−2.93	0.95	−0.972	0.04	−1.762	0.06	1.05	−0.833
1	2021–2022	42,176	128	0.91	−0.40	0.92	0.059	12.77	0.014	0.73	1.04	0.061
1	2022–2023	42,100	73	0.56	−1.05	0.90	0.053	0.62	0.066	0.46	1.02	0.053
1	2023–2024	42,054	139	1.16	−0.38	0.88	0.050	2.08	0.045	0.96	0.98	0.052
1	2024–2025	42,014	126	1.06	−0.53	0.87	0.042	0.91	0.038	0.90	0.99	0.044
3	2018–2021	42,430	570	0.81	−1.46	0.59	−0.056	–	–	0.61	0.74	−0.032
3	2019–2022	42,307	461	0.83	−0.82	0.80	0.079	0.77	0.012	0.65	0.95	0.082
3	2020–2023	42,171	208	0.36	−1.49	0.88	−0.046	0.28	−0.295	0.31	0.99	−0.037
3	2021–2024	42,175	355	0.72	−0.60	0.93	0.075	0.92	0.044	0.63	0.94	0.064
3	2022–2025	42,099	350	0.80	−0.67	0.87	0.054	1.04	0.029	0.73	0.83	0.035
5	2020–2025	42,170	520	0.50	−1.25	0.83	0.046	–	–	0.40	0.97	0.033

Table S9 Calibration in each city's test window (existing conditions only, and all blocks): raw, and after logistic recalibration on out-of-fold training-window probabilities.

City	Blocks	Rate	O/E	Slope	Skill	O/E rec.	Slope rec.	Skill rec.
Los Angeles	E	0.33%	1.29	0.81	0.074	1.17	0.94	0.079
	E+C+Z+P	0.33%	1.59	0.80	0.058	1.24	0.97	0.070
Dallas	E	1.33%	1.01	0.78	0.056	0.96	0.92	0.060
	E+C+Z+P	1.33%	1.18	0.88	0.129	0.97	1.03	0.132
New York	E	0.47%	1.09	0.90	0.034	1.04	0.95	0.035
	E+C+Z	0.47%	1.22	0.91	0.061	1.13	1.00	0.065
Milwaukee	E	0.06%	0.34	0.37	−1.783	0.49	0.80	−0.230
	E+C+Z	0.06%	0.42	0.21	−2.022	0.53	0.71	−0.433
Phoenix	E	0.36%	1.12	0.57	0.038	1.10	0.76	0.048
	E+C+Z+P	0.36%	1.60	0.43	−0.001	1.01	0.79	0.023
San Diego	E	0.07%	0.47	0.36	−0.026	0.46	1.09	0.000
	E+C+Z+P	0.07%	0.78	0.55	−0.057	0.41	1.12	0.000
Chicago	E	0.26%	0.80	0.82	0.000	0.76	1.03	0.005
	E+C+Z	0.26%	0.86	0.87	−0.007	0.75	1.04	0.015

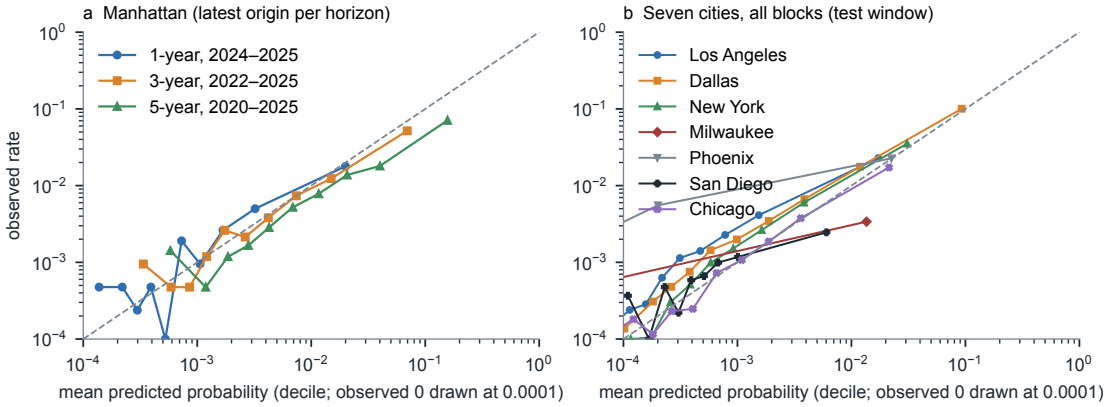


Fig. S2 Reliability of redevelopment probabilities (log axes; dashed line: perfect calibration)

- **Probability × capacity**: $\sum_i p_i K_i$, where p_i is the probability of redevelopment by the end of the window. In the rezoning replays the model's horizon h is extended to the T years to 2025 with a constant hazard, $p_T = 1-(1-p_h)^{T/h}$.
- **Probability × empirical yield**: redeveloped lots seldom reach capacity. In the cross-city test the expected yield of a developed lot is the mean homes added by developed lots of the same zoning class (existing class where zoning is missing) and lot-size bin in the training window (at least 20 lots, else the class, else all lots), so $E[\text{homes}] = \sum_i p_i \hat{y}_i$. In the rezoning replays it is $\sum_i p_{T,i}\, \bar{s}\, K_i$, with $\bar{s}$ the mean share of capacity homes added by soft sites redeveloped in the training pairs.

Probability × capacity and probability × yield are expected values of gross homes added (removals are not subtracted). Their 95% ranges come from 20,000 draws in which each site is redeveloped independently with its probability and, for the yield version, takes a share drawn from the training sites. These ranges are conditional on the model's probabilities: they omit model error and correlation between sites, and are narrower than a full prediction interval.

Build-out proxy. In the cross-city test, zoning and plan build-out are not legal capacity (most cities do not publish maximum FAR as data). Each lot's build-out is the homes per hectare that lots whose existing use is in the class its zoning (or plan) designates reach in the city today (90th percentile among those with homes, at least 30 lots), times its lot area, minus its homes. It measures what a class's dense lots hold today, not what the rules allow.

S6 Optimizer

Objective and constraints. For a plan u in scope S, maximise

$$J(u) = \text{overall}_S(u) - \lambda\,|\{i : u_i \neq u_i^0\}|, \qquad \lambda = 0.0005^\dagger, \tag{S10}$$

subject to at most $B = \max\{1, \lfloor 0.05^\dagger\, n_{\text{eligible}} \rfloor\}$ changes. Eligible parcels exclude landmarks and historic districts, parks, public facilities, transportation and utility lots, and lots without a known use. Allowed uses on a parcel are those its zoning permits (Section S8); homes may only become residential or mixed use; built lots may not become open space, parking or vacant land; and converting an existing building is allowed only if that conversion was observed in Manhattan at that building size or larger (precedent).

First-order gains. For each parcel and use, the change in J is estimated as

$$g_{iu} = \sum_k a_k\,\Delta x_{iuk} + w_{\text{fit}}\,[\log P(u \mid \text{context}) - \log P(u_i^0 \mid \text{context})] - \lambda, \tag{S11}$$

where $a_k = 100\,\delta_k\,\omega_k/|b_k|$ for totals and $100\,\delta_k\,\omega_k$ for shares, ω_k the indicator's weight in the overall score (pillar weight over the number of indicators in the pillar), $w_{\text{fit}} = 0.002^\dagger$, and Δx_{iuk} the change in indicator k:

- exact for the additive indicators (homes, jobs, value, green space, runoff, emissions);
- for access, mix and green cover, the first-order effect on residents within the radius, computed by convolution on the raster: added open space for residents lacking a park, added floor area for residents in under-served cells, residents added to under-served cells (dilution), the entropy gradient $-(\log s_g + H \log 5)/(T \log 5)$ of land-group shares s_g for mix, and the open-space share gradient for green cover;
- plus the change in who is counted: a parcel's added residents enter each resident-weighted mean with that parcel's own value.

The gains are exact for additive indicators and first order for the rest, which saturate.

Search. Iterated best response over $3^\dagger$ rounds: in round r the $\lceil (B - \text{changes so far})/(R - r) \rceil$ parcels with the largest positive gain switch to their best use, then access and gains are recomputed. Because J is the potential function of the game in which each parcel chooses its use, each round can only improve the estimate. Polishing re-scores exactly the $30^\dagger$ best moves (add a move, swap it with the weakest change, or drop a change) for up to two passes. The Studio uses polishing; the candidate plans for votes do not. The result is the best plan found, not a proven optimum.

Likelihood layer. When on, candidates are restricted to the fifth of parcels with the highest five-year probability of redevelopment. Probabilities restrict the candidate set; they do not weight moves.

Exact search. The same objective J is stated as a mixed-integer linear program over binary choices x_q, one per pair $q = (i, u)$ of an eligible parcel and a new use, with $\sum_u x_{iu} \leq 1$ and $\sum_q x_q \leq B$, and solved by branch-and-bound with HiGHS (Huangfu & Hall, 2018). Every term is written as a change from the start plan.

- **Totals** are linear in x; their ± 100 score clip needs one binary where the lower clip can bind.
- **Shares.** The five shares are resident-weighted means with one denominator, the scope's residents D, and lie in $[0, 1]$, so their scores never reach the clip. They enter as one weighted mean $\tau = Q/D$ of the cells' composite access $A_c = \sum_k 100\,\omega_k a_{kc}$, or one per pillar for trade-offs. With $\tau = \tau_0 + \delta$ and $A_c = A_c^0 + \alpha_c$,

$$\delta D = N' = \sum_c R_c^0\,\alpha_c + \sum_q \Delta r_q\,x_q\,(A_{c(q)}^0 - \tau_0) + \sum_q \Delta r_q\,x_q\,\alpha_{c(q)}, \tag{S12}$$

  where R_c^0 are a cell's residents and Δr_q the residents pair q adds. Products of a binary and a bounded variable are replaced by McCormick envelopes (McCormick, 1976), which are exact at binary points. D is split into intervals of ratio 1.05; within $[D_l, D_h]$, $\delta \leq N'/D_l$ when $\delta \geq 0$ and $\delta \leq N'/D_h$ otherwise (one binary).
- **Access per 50 m cell**, from sums within the radius (row prefix sums):
  - park access is a binary y_c with open area within the walk at least the park minimum times y_c, a covering constraint (Church & ReVelle, 1974), and a cover inequality that links y_c to the open space added nearby;

Table S10 NSGA-III against the true front: median (minimum) over blocks of the hypervolume share and of the share of true-front plans returned; random search with the same number of evaluations for comparison.

Blocks	n	Hypervolume share	Front plans found	Random search
Small (exact)	10	1.000 (1.000)	1.000 (0.846)	1.000 (1.000)
Medium (first order)	10	1.000 (0.986)	0.962 (0.638)	0.964 (0.944)
Medium, re-scored	10	1.000 (0.991)	–	–

  - services and facilities satisfy $(a^0 + \alpha)\, t\, (R^0 + \Delta R) \leq F^0 + \Delta F$ and $a^0 + \alpha \leq 1$, with residents R and floor area F within the radius;
  - green cover satisfies $g\, T \leq L_{\text{open}}$, for grouped land T and open land L_{open};
  - land-use mix satisfies Gibbs' inequality, $H\, T \log 5 \leq \sum_g L_g\, (-\log \bar{s}_g)$ for any shares $\bar{s}$: a tangent of the entropy, exact when $\bar{s}$ are the cell's shares.
- **Products** of an access change and the resident or land change within its radius are exact for the pairs with the largest changes near the cell and the pairs a solution uses, and bounded by McCormick envelopes, split by sign, for the rest. Ranges come from budget-limited extremes and the residents interval; for mix, from the Fannes–Audenaert continuity bound of entropy (Audenaert, 2007).

Every relaxation over-estimates J, so the solver's bound is an upper bound on J over all feasible plans. After each solve the plan is re-scored by the engine, exact products and entropy tangents are added at the plan, and the loop stops when the bound and the best exact score agree within a relative 10^{-4} (certified optimal) or at the time limit (a certified gap). With a plan fixed, the model's value equals the engine's score (a test of the formulation).

Exact trade-offs. Optimal plans for pillar weights on a simplex grid (step 0.25), the plan that maximises the weakest pillar, and the augmented ε-constraint method AUGMECON2 (Mavrotas & Florios, 2013) give the efficiency frontier (Polasky et al., 2008). Weighted optima are supported Pareto plans; the ε-constraint plans add unsupported ones. Pillar scores are real-valued, so the ε-constraint plans are a representation of the front, not a guaranteed complete Pareto set (AUGMECON2 is exact for integer objective values with unit steps). Trade-off rates are the change in one pillar per point of another between neighbouring optimal plans along each edge of the weight simplex. Today's land use scores zero on every pillar, so the weakest pillar of the balanced plan is how much every pillar can rise at once: the efficiency gap, a distance to the frontier in the direction of equal improvement. Robustness is the share of frontier plans that stay non-dominated when re-scored under the low and high coefficient variants.

GNN policy. A graph neural network proposes a plan at once and starts the exact search, in the manner of neural diving (Nair et al., 2020). It has 3 GraphSAGE layers (Hamilton et al., 2017) (hidden size 64) over the parcel graph (adjacent, near and same-zoning edges) of an area's study area. Each parcel's inputs are its lot and floor area, residents, current use, the redevelopment model's probability and likely new use, the pattern model's use fit, and, for the area, whether it is selected, which uses are allowed, the pillar weights, the budget share and the first-order gain of each use under those weights. Its output is logits over "stay" and the nine uses. It was trained by cross-entropy (changes weighted five times) on the certified optima of 180 random areas in districts 101–108 (150–300 m around a random parcel, a budget of 5–15% of eligible parcels, pillar weights uniform on the simplex) and tested on 80 areas in districts 109–112. A plan is the most likely changes up to the budget.

NSGA-III explorer. For a selection of parcels, three objectives (the first-order social, economic and environmental gains) are searched by NSGA-III (Deb & Jain, 2014) in pymoo: 45 Das–Dennis reference directions (eight partitions), population 92, 120 generations, simulated binary crossover (probability 0.9, $\eta = 15$) and polynomial mutation ($\eta = 20$) with integer rounding, a fixed seed. Every plan evaluated is archived; the non-dominated plans of the archive are re-scored exactly, plans that another re-scored plan beats on every pillar are dropped, and at most 24 plans are shown (the best per pillar, then evenly spaced by total gain).

Test of the explorer. On Manhattan tax blocks drawn at random, every combination of options was enumerated (Table S10). On 10 small blocks (64–1,024 plans) every plan was also scored exactly; on 10 medium blocks (24,576–327,680 plans) the first-order front is known. Quality is the hypervolume of a front as a share of that of the true front.

A defect found by this test. The first version of the first-order model omitted mix and green cover and the change in who is counted. On small blocks its rank correlation with exact pillar scores fell to -0.35 (environmental), and on one block the explorer's front covered 0.5% of the exact front's hypervolume. With the terms above, the rank correlation is at least 0.99 on every pillar.

Table S11 Indicator weights of the five stakeholder agents (assumptions; each sums to one).

Agent	Homes	Jobs	Value	Park	Services	Facilities	Mix	Green sp.	Runoff	Emissions	Green cov.
Developer	0.35	0.20	0.45								
Residents' association	0.15			0.25	0.20	0.20	0.10				0.10
Environmental advocate								0.30	0.20	0.30	0.20
City budget office	0.20	0.25	0.55								
Local business alliance		0.40			0.35		0.25				

Table S12 Disagreement among the nine rules by candidate set (12 districts): median candidates, median and maximum number of distinct winners, districts where all rules agree and with a Condorcet winner.

Candidate set	Candidates	Winners	Max	All agree	Condorcet
generated	19	4	5	0	7
ideal plans only	6	3	4	0	11
generated + random	35	4	5	0	7
random only	17	2	4	4	11

With each block's resident weights drawn independently, the residents' winner is unchanged in a median 100% of draws.

S7 Decision module

Stakeholders and utilities. Five agents stand for parties that commonly testify or decide at land-use hearings: developers (the applicant), residents (community boards and civic associations), environmental advocates, the city's budget office, and local business associations. Each agent's utility for plan a is $U_{va} = \sum_k w_{vk}\,\text{score}_k(a)$ with the weights of Table S11, so the status quo is worth zero to everyone. The weights are editable assumptions, not elicited preferences. Each agent holds one seat. One agent per interest does not represent the variety within it; the residents' referendum below and the sensitivity tests show how much that matters.

Candidate plans. The status quo; each agent's ideal plan (the optimizer with that agent's weights); ten pairwise compromises (the optimizer with the mean of two agents' weights); a balanced plan (equal pillars); a consensus plan (only moves that improve every agent's first-order utility); and a parcel-by-parcel ballot (on each eligible parcel the council ranks the allowed uses by weighted Borda and a change passes with an absolute majority of seats). All use the same change budget. Identical plans are merged, and every candidate is re-scored exactly.

Rules. With weighted voters and ties allowed: plurality; Borda; instant runoff; Copeland; Schulze (the headline rule: Condorcet-consistent, monotone and independent of clones); Kemeny–Young (exact integer program, 10 s limit); approval (plans better than the status quo); score (utilities rescaled to 0–10 per voter); and majority judgment (seven grades from utility). Ties are broken by Borda score. Also reported: the Condorcet winner and loser, the Smith set, whether one voter can change the Schulze result by misreporting, and the Nash bargaining solution with the status quo as disagreement point.

Residents' referendum. Today's residents vote by tax block, weighted by population. A block's utility for a plan is the residents' weights applied to the change in its own access, mix and green cover, plus homes in the study area; differences below $0.01^{\dagger}$ points are ties (just-noticeable difference, an assumption). Future residents and other areas do not vote.

Robustness and sensitivity. The winner's Schulze win probability is estimated over $200^{\dagger}$ draws in which each indicator's delta is drawn from a triangular distribution over its sensitivity range (one quantile per indicator, shared by all plans, because a coefficient that is higher than assumed is higher for every plan) and seats from a Dirichlet distribution centred on the seats (concentration $50^{\dagger}$). The sensitivity tests redraw each agent's weights (Dirichlet, concentration 20), add random feasible plans to the ballot, sweep the threshold, triple each agent's seats in turn, and draw each block's resident weights independently.

S8 Zoning and access as represented

- **Represented**: which of the nine use groups each zoning district permits (simplified rules for New York; crosswalks of district classes elsewhere), maximum FAR by residential, commercial and community-facility use where the city publishes it, landmarks and historic districts as protections.

- **Not represented**: height and setback limits, lot coverage, parking requirements, inclusionary and other bonuses, special-district rules, discretionary approvals and building codes. A plan that passes the screen may still be infeasible.
- **Access** uses Euclidean disks on a raster, not walking routes, so barriers such as highways, rail yards and rivers are ignored and access near them may be overstated.

S9 Additional results

Tables S13–S16 give the cross-city panels and their out-of-time results, referred to in the main text.

Table S13 Test panels. Models are fitted on the training window and scored on the test window.

City	Development	Source	Train	Test	Lots	Developed	Rate
Los Angeles	new building or floor area +25%	county assessor rolls	2021–2023	2023–2025	667,844	2,215	0.33%
Dallas	new building or floor area +25%	appraisal district files	2022–2024	2024–2026	291,841	3,892	1.33%
New York	new building or floor area +25%	MapPLUTO	2021–2023	2023–2025	856,901	4,064	0.47%
Milwaukee	new building	MPROP	2021–2023	2023–2025	147,539	85	0.06%
Phoenix	change of use group	MAG land-use inventory	2022–2024	2024–2026	503,635	1,835	0.36%
San Diego	change of use group	SANDAG land-use inventory	2023–2024	2024–2025	272,281	202	0.07%
Chicago	new-construction permit	building permits	2021–2023	2023–2025	606,294	1,548	0.26%

Lots already under construction at the start of a window are left out. Floor area +25% also requires +186 m^2 (2,000 sq ft). Change of use group: the lot changes to a built use of another of the nine groups, or gains homes.

Table S14 Where development happens: out-of-time AUC by block set. E existing conditions, C neighbourhood context, Z zoning, P adopted plan; – where the city has no such layer.

City	E	E+C	E+C+Z	E+C+P	E+C+Z+P	Best set (any)
Los Angeles	0.863	0.889	0.894	0.891	0.894	E+C+Z
Dallas	0.897	0.917	0.921	0.919	0.922	E+C+Z+P
New York	0.892	0.914	0.919	–	–	E+C+Z
Milwaukee	0.911	0.896	0.924	–	–	E+Z
Phoenix	0.859	0.793	0.842	0.847	0.853	E+Z+P
San Diego	0.571	0.687	0.684	0.707	0.703	E+C+P
Chicago	0.824	0.888	0.891	–	–	E+C+Z

Best set: the highest AUC among all fitted combinations, including those without context.

Table S15 Transfer: AUC of a model fitted on the other cities' training windows / the city's own model, on the city's test window.

City	E	E+C	E+C+Z	E+C+Z+P
Los Angeles	0.726 / 0.863	0.822 / 0.889	0.827 / 0.894	0.828 / 0.894
Dallas	0.845 / 0.897	0.873 / 0.917	0.880 / 0.921	0.872 / 0.922
New York	0.824 / 0.892	0.871 / 0.914	0.874 / 0.919	0.865 / 0.919
Milwaukee	0.958 / 0.911	0.966 / 0.896	0.966 / 0.924	0.966 / 0.924
Chicago	0.736 / 0.824	0.805 / 0.888	0.800 / 0.891	0.814 / 0.891

Cities whose outcome is a building or a permit. A missing plan layer is coded as unknown.

Table S16 Out-of-time average precision and precision among the top 1% of lots.

City	Base rate	AP, E	AP, all	All blocks	Top 1%, E	Top 1%, all
Los Angeles	0.33%	0.132	0.143	E+C+Z+P	11.2%	12.7%
Dallas	1.33%	0.108	0.232	E+C+Z+P	11.8%	29.3%
New York	0.47%	0.073	0.122	E+C+Z	11.1%	15.1%
Milwaukee	0.06%	0.024	0.031	E+C+Z	2.3%	2.8%
Phoenix	0.36%	0.091	0.054	E+C+Z+P	9.7%	8.4%
San Diego	0.07%	0.002	0.002	E+C+Z+P	0.7%	0.6%
Chicago	0.26%	0.013	0.043	E+C+Z	2.4%	6.1%

S10 Reproducibility

The code (MIT licence), the result file behind every number, and a manifest of every input (URL, retrieval date, SHA-256) are in the repository, anonymised for review at `https://anonymous.4open.science/r/planarena-cus`.

`make reproduce` rebuilds the data and every result from public sources, and `python -I -m planarena.paper` rebuilds the paper and this document from the result files.

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